

Lomonosov Moscow State University
Faculty of Economics
Department of the History of the National Economy and Economic
Thought



Graduate qualification thesis (Bachelor's thesis)

The Impact of Mining Costs on the Cryptocurrency Market

Written by a student of group e401

Surov Timofei Olegovich

Academic supervisor

Associate Professor Rozinskaya Natalia Anatolievna

Moscow, 2023

Contents

1	Introduction	2
1.1	Relevance	2
1.2	Literature review	2
1.3	Aims, objectives, and methods of the research	4
1.4	Novelty	5
2	Theoretical foundations of the cryptocurrency market	6
2.1	The theory of money	6
2.2	The Austrian theory of private money	7
2.3	The model of private electronic money	9
2.4	Cryptocurrency as a type of private electronic money	15
2.5	The model of demand for cryptocurrency	16
3	Theoretical aspects of the costs of issuing a cryptocurrency	19
3.1	The environmental aspect	19
3.2	The cost-of-production model of cryptocurrency pricing	20
3.3	The endogenous model of the cryptocurrency market	23
4	An empirical model of the impact of costs on the cryptocurrency market	24
4.1	Formulation of the hypotheses	24
4.2	Data	25
4.3	Model selection	29
4.4	Model specification	31
4.5	Model results	32
5	Conclusion	35

1 Introduction

1.1 Relevance

Cryptocurrencies have become a true breakthrough in global science and economics. Today, everyone has heard of this kind of money. A cryptocurrency is not only a means of payment but also an actively used financial asset. In 2008, the proposal to create a decentralized electronic currency turned out to be not a phantom idea of the proponents of private money, but a real revolution in the world of money. The idea of money whose supply is absolutely independent of state and supra-state bodies, and even of its own creators, undoubtedly won their support. In 2010 the first transaction using Bitcoin took place, and it became clear that the work of S. Nakamoto [17], which described the mechanism of the first cryptocurrency, goes far beyond theoretical investigation.

There are many works that study the price of cryptocurrencies by analogy with other financial assets/currencies. However, taking into account the structure of the costs of issuing a cryptocurrency (mostly energy costs), the question of the environmental friendliness of a particular cryptocurrency arises. Proposals have been put forward on how, according to the characteristics of the energy costs of issuing a particular cryptocurrency, they can be divided into “clean” and “dirty” [20]. It is logical to assume that in today’s world, where the environmental agenda is one of the foremost global issues, the energy “efficiency” of a cryptocurrency affects the intensity of its mining and its price in particular.

Interest in studying the costs of cryptocurrencies was fueled by the incident in Svaneti, Georgia. The government introduced a program of providing free electricity to support remote mountain communities and to stimulate tourism in the region. This program attracted an enormous number of “miners” from all over the country and beyond, since, by using free electricity, one can significantly increase the profitability of a mining farm. This led to a chronic electricity shortage in the region and the subsequent protests against the “miners.” This case shows that costs are a key factor in the decision to issue a cryptocurrency, and that the issuance of a cryptocurrency (“mining”) itself has high territorial mobility.

1.2 Literature review

The bibliography can be roughly classified into four categories:

1. Theoretical works related to the theory of money, private money, and the theory of cryp-

tocurrencies ([1], [2], [4], [6], [9], [14], [15], [16], [17], [23], [24], [25])

2. Environmental studies of the costs of “mining” and their impact on the environment ([19], [20])
3. Empirical studies analyzing the cryptocurrency market ([3], [5], [10], [12], [13], [18], [21])
4. Theoretical works related to the empirical methods used in this research ([7], [8], [11], [22]).

The works of the first category constitute the theoretical foundation for empirical research in the field of cryptocurrencies and for models of money circulation. My research relies on the general theory of money, the works of authors of the Austrian school and the neo-Austrians on private money, and contemporary research in the field of private electronic money and cryptocurrencies.

The works of the second category closely examine the question of the costs of “mining.” Since the costs of issuing a cryptocurrency are also specific to my research, I make use of the definitions and structural analysis of costs presented in the environmental articles I have listed.

The works of the third category represent the most significant directions of empirical research: models of demand for cryptocurrency, models of the “fair” price of cryptocurrencies based on their costs, and structural models of the cryptocurrency market that include the costs of their issuance. My work is related to the last two, but it differs significantly in methodology and theoretical approach from the articles I have listed.

The works of the fourth category contain the tools necessary for conducting empirical research.

Based on the literature review, the theoretical foundation of cryptocurrencies and the environmental approach to the costs of their issuance have already been built, and the main directions of empirical research have been identified. Modern approaches to empirical research assume a significant impact of issuance costs on the price of a cryptocurrency. Some approaches hold that there is no feedback, while other authors argue that the price of a cryptocurrency and the aggregate costs of its issuance affect each other bidirectionally.

My work complements the theoretical analysis of the “mining” costs of a cryptocurrency by singling out the energy/environmental aspect of the costs as the key one in a quantitative study of the impact on the cryptocurrency market, and it also complements the empirical research on the relationship between the cryptocurrency market and “mining” costs.

1.3 Aims, objectives, and methods of the research

The aim of this research is to determine the impact of the factors that determine the difficulty and cost of mining on the price of a cryptocurrency.

To achieve this aim, a number of objectives must be accomplished:

- Identify the main differences between the cryptocurrency market and the markets for other financial assets
- Analyze the cost structure of “mining”
- Analyze the existing contemporary scientific literature, as well as models related to the cryptocurrency and electricity markets
- Collect data on the prices and issuance costs of the cryptocurrency under study
- Analyze the dynamics of cryptocurrency price changes
- Estimate the impact of “mining” costs on the price of a cryptocurrency

The object of the research is the cryptocurrency market.

The subject of the research is the characteristics of the “mining” costs of Bitcoin and altcoins.

The hypotheses of the research are intended to answer the question of whether there is a significant bidirectional relationship between the price of a cryptocurrency and the aggregate costs of its issuance in the short run (up to 2 weeks) and the long run (more than 2 weeks). The expected result: the presence of a significant positive bidirectional relationship between the indicators in the short run, possibly the presence of a long-run one-directional impact of the cryptocurrency price on the aggregate “mining” costs, and the absence of an impact of the aggregate costs on the cryptocurrency price in the long run.

The methods used to achieve the aim of the research are a theoretical analysis of scientific approaches to cryptocurrency costs, as well as regression analysis to obtain quantitative estimates of the impact of factors related to cryptocurrency issuance.

The data used to conduct the research, on the price of Bitcoin and the consumption of crypto-farms issuing Bitcoin, are daily, from January 1, 2015 to December 4, 2021.

The information base of the research: textbooks and methodological aids, scientific literature from technological, economic, historical, and environmental scientific publications, and Internet websites.

1.4 Novelty

My research makes the following scientific contribution to the academic literature on this topic:

- An approach to analyzing the impact of the aggregate “mining” costs on the cryptocurrency market in terms of the electricity expended on its issuance.
- Proof of a bidirectional cause-and-effect relationship between the aggregate costs of issuing a cryptocurrency and the price of a cryptocurrency. A refutation of the one-directional impact of costs on price.
- The use of a different approach (the SVAR model) to estimating the impact of “mining” costs on the cryptocurrency market (as well as the feedback).
- The separation of quantitative estimates into estimates of the short-run and long-run impact of the variables on each other.
- An update of the quantitative estimates.

2 Theoretical foundations of the cryptocurrency market

It should be noted that the theoretical foundations of cryptocurrency analysis are deeply rooted in the history of economic thought and in the views of economists of various schools on the global monetary system. Although cryptocurrencies do not have the legal status of money, in countries such as Germany and Japan they are an officially permitted means of payment. Cryptocurrencies in fact fit well under the definition of private money, about which a great deal has been written by renowned authors. Therefore, it is impossible to discuss cryptocurrencies in isolation from, for example, the Austrian and neo-Austrian ideas of private money or the theory of private electronic money.

2.1 The theory of money

According to the generally accepted definition, money is a universal equivalent that serves as a measure of the value of goods or services and is easily exchanged for them. The theory of money begins with the metallist theory of money. Within this theory it was assumed that the value of money is determined by the value of the material from which it is produced. The subsequent nominalist theory rejected the metallist concept, and the value of money was determined by the face value assigned to it. Thanks to this theory, the existence of paper money became possible. According to the Marxist theory of money, money is merely the value expression of the labor expended on producing a good; the necessity of money is explained by the simplification of exchanging one good for another, since money commensurates the value of all goods. Cryptocurrencies can quite definitely be regarded as money in many theories of money, since they satisfy the definition and perform the function of a means of payment assigned to money. However, more important for understanding the characteristics of cryptocurrencies is, of course, the theory of private money, owing to the specifics of cryptocurrency issuance.

The history of the distinct idea of non-state money in economic science begins in the second half of the twentieth century in the works of authors of the Austrian economic school, despite the fact that such a model had been practiced before. This is because no one in economics had proved that a model with state money is more efficient than one with private issuers. Before the studies of the Austrians, most scholars attached no importance to who the issuer of money was, or agreed that the issuer should be the state. Most likely, this is explained by the fact that pre-Keynesian scholars often did not assign a significant role to money. An exception is the views of H. Thornton [24], who criticized A. Smith's "real bills" doctrine, which assumed commodity collateral for paper money.

Although in the era of the gold standard money was nonetheless state money in many countries, there was no clear division into state and private money in pre-Keynesian works. In works describing general equilibrium in the economy, the general supply of money was studied regardless of who issues that money. Thus, Kondratiev held that the money supply under the gold standard depends on the profitability of gold mines, which is determined by market pricing [1]. Under such conditions it makes no difference who carries out the issuance, since the money supply is endogenous, and it can be affected only through influencing the price of gold. In the Keynesian era, the most popular idea became that of a central bank that carries out the issuance of money and/or securities and is able to influence the money market. In some regions, despite this, the money supply was determined not necessarily by a single, and not necessarily a state, body. Thus, for example, in Hong Kong, Northern Ireland, and Scotland several issuing banks operated (and still operate), although they are subject to the influence of a higher authority (the Hong Kong Monetary Authority and the Bank of England). However, in the work of the post-Keynesians the idea of the endogeneity of the money supply appears, that is, the money supply is subject to the laws of supply and demand, and the role of the central bank is not as significant as in Keynes, because in the model the money supply is not exogenous [14].

I note that Kondratiev's concept is very remarkable for this research, since it is essentially a model of money pricing that includes the factor of issuance costs. One cannot say that cryptocurrencies are similar to money under the gold standard; however, the idea of issuing money only if it is profitable for the issuer lies at the foundation of cryptocurrencies.

2.2 The Austrian theory of private money

With the collapse of the Bretton Woods system, the contradictions between the supporters and opponents of a powerful central bank grew, and the universal idea of private money arose. The Austrian school was the most ardent critic of the state monopoly on money and was always quite radical in its ideas. Soon after the abolition of the gold standard, the Austrians proposed restoring it, but not by returning to the previous state of the monetary system, rather by reforming it as a whole and abandoning the state monopoly on money. In his work "The Denationalisation of Money" [9], Hayek emphasized that, for economic efficiency, money must be subordinated to the market mechanism, since the state monopolist does not have time to react to fluctuations in the demand for money and investment. The proposal to return to the gold standard is explained by the fact that the gold standard is a guarantee against the impossibility of private over-issuance and the inflation that follows from it. Hayek was not enthusiastic about

the gold standard, but he tolerated its shortcomings for the reason stated above. After the publication of Hayek’s work, it was precisely this work that became the main conception of private money held by researchers of the late twentieth century. In “Denationalisation” he did not take into account two facts: the importance of network effects, and the fact that currencies with low demand need not necessarily leave the market. As it would turn out in the twenty-first century, private money is possible even without a gold standard.

On the basis of Hayek’s works, an explicit parallel can be drawn between the theory of the last century and the reality of the present. It is interesting that many authors highlight the connection between the ideas behind cryptocurrencies (Bitcoin in particular) and the works of F. Hayek ([2], [4]).

In the work of D.S. Bas [4], the commonality between Hayek’s philosophy and the philosophy behind cryptocurrency is highlighted: the state is alienated from the control of cryptocurrency. In addition, the use of cryptocurrencies, like money in “The Denationalisation of Money,” depends entirely on the acceptance of a particular currency by economic agents. Third, Hayek called for creating a system in which monetary policy either could not be conducted or would have no effect. Owing to the algorithm of cryptocurrency issuance, monetary policy is impossible in the cryptocurrency market. Finally, in “Denationalisation” Hayek advances the idea that confidence in the issuer should stem from the stability of the market system, not from the solvency of the central bank or other state bodies. Decentralized cryptocurrencies rely entirely on this idea [3].

D.S. Bas [4] and others, in their article, move on to studying the development of cryptocurrencies, tracing their roots back to the creation of Bitcoin in 2009. The authors argue that cryptocurrencies represent a new form of decentralized currency that is not subject to the control of any central authority. This, in their view, corresponds to Hayek’s vision of a competitive currency market. They point out that Hayek argued that centralized planning is doomed to failure because it cannot account for the complexity of the market, and that competition is necessary to ensure the efficient allocation of resources. Accordingly, if humanity cannot completely abandon state money, cryptocurrencies can increase the efficiency of the global monetary system by acting as a kind of mediator-liquidator of information asymmetry. The article also examines some of the problems that cryptocurrencies face, such as their volatility and lack of widespread adoption. However, the authors argue that these problems are not insurmountable and that cryptocurrencies have the potential to change our relationship with money. Overall, the article presents a compelling argument for the relevance of Hayek’s ideas in the context of the cryptocurrency revolution. It suggests that cryptocurrencies are a natural continuation of Hayek’s vision of decentralized decision-making and competitive markets, and that they have

the potential to change our relationship with money and finance.

The problems facing cryptocurrencies are examined in more detail in the work of F. Ametrano and others [2]. The authors propose a new approach to achieving price stability in cryptocurrencies. The authors argue that the current volatility of cryptocurrencies is a serious obstacle to their widespread use as a medium of exchange. They propose a system called “Hayek money,” which would use an algorithmic approach to adjust the supply of a stablecoin in response to changes in demand. The “Hayek money” system would be based on a stablecoin pegged to a basket of goods and services, rather than to a single currency such as the US dollar. The stablecoin would be backed by a reserve of other cryptocurrencies, which would be used to adjust the supply of the stablecoin in response to changes in demand. The algorithm used to adjust the supply would be based on the principles of the Austrian school of economics, which emphasizes the importance of market signals and decentralized decision-making. The authors argue that such an approach would provide several advantages over existing stablecoin systems, such as Tether and USDC. First, it would be more resistant to manipulation and fraud, since the reserve would consist of several cryptocurrencies rather than just one. Second, it would be more transparent and verifiable, since the algorithm used to adjust the supply would be open and subject to public scrutiny. Finally, it would be more adaptive to changes in the market, since the algorithm would dynamically respond to changes in demand rather than rely on fixed rules.

However, in my view, such a stablecoin may still face obstacles in the form of the possibility of market manipulation, as well as an unstable reserve (although the proposed reserve option would presumably be more stable than that of existing stablecoins). I believe that it is precisely an analysis of cryptocurrency supply from the perspective of costs that would help to study the stability of the reserve, since, under conditions of decentralized private issuance (“mining”) of a cryptocurrency, the collective decision of individuals about issuance is made on the basis of calculating the profitability of “mining.” If one approaches this question from this side, one can predict fluctuations in supply on the basis of expected shocks in the markets for electricity, technical equipment for cryptocurrency mining, newly developed “mining” algorithms, and other kinds of costs.

2.3 The model of private electronic money

It will also be relevant for this research to consider models of private electronic money, which are the ideological continuation of the Austrian models of private money. Electronic private money is a format of money that includes, for example, such Russian settlement services as WebMoney

and Qiwi. The main distinguishing features of electronic money are very high liquidity and the convenience of settlements. The assumptions of the theoretical models are quite adequate for a world in which cryptocurrencies are actively used as a means of payment (which is the direction in which the global economy is moving). Thanks to this, on the basis of models of private electronic money, a number of conclusions about the cryptocurrency market can be drawn.

I will consider the model of demand for private electronic money presented in the work of J. Chiu and T. Wong, “E-money: Efficiency, stability and optimal policy” [6]. The main property of electronic money used in the model is its high liquidity. The model is an extension of the classical model of ordinary private money and is a kind of superstructure over the everyday banking system with state money, which exists in practice in most countries of the world. The full diagram of the model is presented in Fig. 1

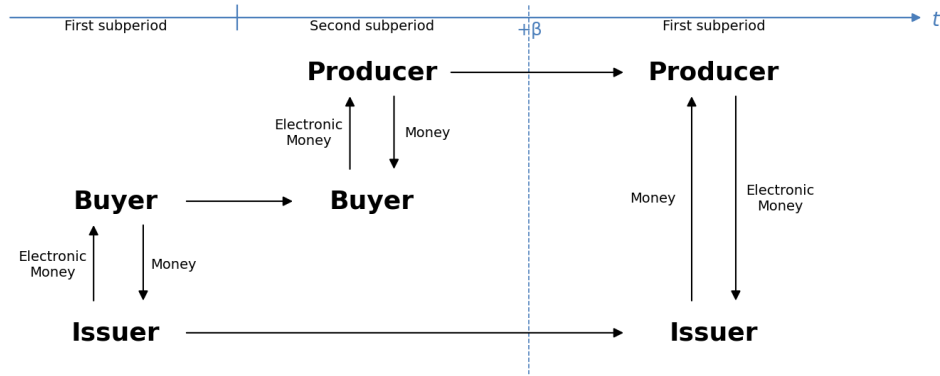


Figure 1: Diagram of the private electronic money model. (Source: compiled by the author)

First of all, let us define the environment. There is an unlimited number of periods ($t = 0, 1, 2, \dots$) and two subperiods. Throughout the entire horizon, three types of economic agents act in the economy in different subperiods: let there be a monopolist firm (the issuer) that engages in the issuance of electronic money, and the shares of this firm belong to the other economic agents in the model – producers and buyers. The model is oriented toward the demand for money, that is, it emphasizes the decision made by the consumer, who uses money as a means of payment and accumulation. Let us consider his utility function (expression 1).

$$U = \sum_{t=0}^{\infty} \beta^t (-y_t + u(q_t)), \quad \beta \in (0, 1), \quad y_t \in \mathbb{R}_+, \quad q_t \in \mathbb{R}_+. \quad (1)$$

1: Buyer's utility

y_t is the quantity of good 1 that the buyer possesses (most resembling labor), which he

can exchange with the producer for q_t – a certain quantity of good 2, which the entrepreneur produces using good 1. The exchange is carried out either using low-liquidity money or using high-liquidity electronic money. As can be seen, the system represents a complicated variant of the infinite Arrow-Debreu model: a per-period exchange of two goods between producer and consumer using an intermediary – the issuer. It is clear that, by excluding the issuer from the model, we obtain a model of commodity circulation.

The idea in the model is as follows: consumers and producers can either exchange goods using ordinary money, but with a time lag due to its low liquidity relative to electronic money and bearing the costs associated with storing it, or use electronic money, paying a certain fee to the issuer.

$$\mathbb{I}_c = \begin{cases} 1, & \beta u(q) - d \geq \kappa_C \\ 0, & \beta u(q) - d < \kappa_C \end{cases} . \quad (2)$$

2: Condition for choosing electronic money

The model assumes that individuals can use both electronic money and ordinary money as a means of payment. The indicator reflected in system 2 indicates whether individuals are willing to resort to using electronic money: if $\mathbb{I}_c$ equals zero, then the individual is inclined to use electronic money as a means of payment; if it equals one, the individual conservatively pays with ordinary state money. Let us consider the methodology for choosing the condition of this decision. d is the amount of electronic money expressed in ordinary money, $u(q)$ is the utility the buyer derives from consuming q units of good 2, expressed in ordinary money, κ_C reflects the costs of using ordinary money. In essence, the left-hand side of the condition reflects the intertemporal costs of using electronic money, the right-hand side – those of using ordinary money. Since in the model electronic money can be only a means of payment, the costs of using it are easy to formulate, whereas the role of ordinary money is somewhat broader: it can be used both in settlements and serve as a means of accumulation for subsequent exchange into electronic money. Therefore κ_C includes storage costs and the so-called “shoe-leather costs.”

$$\mathbb{I}_e = \begin{cases} 1, & \beta u(q) - d - \beta \alpha_B(u(q) - d) \geq \kappa_C \\ 0, & \beta u(q) - d - \beta \alpha_B(u(q) - d) < \kappa_C \end{cases} . \quad (3)$$

3: Condition for combining money and electronic money

As can be understood from the previous paragraph, under certain conditions individuals

may find it necessary to use both ordinary and electronic money as a means of payment. The possibility of combining means of payment is reflected in system 3. $\mathbb{I}_e$ equal to one means that agents use both ordinary and electronic money as a means of payment; equal to zero – only electronic money (that is, they exchange all ordinary money for electronic money). $\beta\alpha_B$ is the probability that the buyer purchases good 2. Thus, the expression contains the buyer's costs (the first two terms on the left-hand side of the condition) and the producer's costs (the third term on the left-hand side of the condition), which must be less than the costs from the time lags and from storing ordinary money. It must be noted that condition 2 was exclusively for buyers and considered their costs, since it is precisely they who store money, whereas not all sellers are willing to conduct settlements in electronic money. It is for this reason that the need may arise to conduct part of the operations in ordinary money.

It should be noted that, in this formulation of the model, each agent's willingness to use money may also be caused by the fact that all the profit received by the issuer is redistributed among them at the end of the period, although the fee rates for producers and buyers in the model may differ. And indeed, in this model, under the condition of maximizing the utilities of buyers and producers, as well as maximizing the issuer's profit, a stationary equilibrium is possible, and when any agent deviates from it, it does not become advantageous for the other agents to deviate. This is explained by the fact that in the model with a monopolist issuer there is only one stationary dominant equilibrium, and since the decision to use electronic money is made jointly by buyers and producers taking into account each other's utilities, if the decision has been made, then the maximum utility (or profit) for each agent is at equilibrium.

The authors separately single out an analysis of the impact of private electronic money on social welfare (within the model – as in the model of ordinary private money – the welfare of all economic agents). Remarkable is the case where ordinary and electronic money are combined as a means of payment, that is, where some sellers do not agree to conduct settlements in electronic money. It turns out to be an interesting fact that the welfare of society, according to the authors' estimates, is potentially higher than in situations where there is no electronic money in the economy, or where all settlements are conducted in electronic money.

Let us consider in detail the impact of each situation on social welfare in the model (see Table 1). We single out four indicators by which the impact on welfare can be grouped: surplus (the agent's intertemporal utility), liquidity of funds (the impact of the liquidity constraint on the equilibrium), externalities, and the monopolist-issuer effect (the impact of the issuer firm's deadweight loss on total welfare). The case $\mathbb{I}_e = 1$ – where only ordinary money is used – is considered the baseline (which is why each element in the table equals zero), and the other two cases are compared with it.

Table 1: Impact on welfare in the private electronic money model. (Source: compiled by the author)

Agent	Indicator	$\mathbb{I}_c = 1$	$\mathbb{I}_e = 1$	$\mathbb{I}_e = 0$
Buyer	Surplus	0	+	0
	Liquidity of funds	0	+	0
	Externalities	0	?	-
	Monopolist-issuer effect	0	-	-
Total		0	+	-
Producer	Surplus	0	0	0
	Liquidity of funds	0	0	0
	Externalities	0	?	-
	Monopolist-issuer effect	0	-	-
Total		0	-	-

$\mathbb{I}_e = 1$ – the case of combining ordinary and electronic money is characterized by a noticeable increase in surplus and a reduction of the liquidity constraint for buyers, and by the absence of an effect on these indicators for sellers, since they spend all their surplus on paying the issuer’s fee. The authors find it difficult to give an estimate of the externalities in this case, but they confirm the presence of deadweight losses as a result of the monopolistic activity of the issuer firm. As a result, one can say that the case is most likely favorable for buyers and unfavorable for sellers, the main drawback being the monopolist’s deadweight loss, which can be eliminated.

$\mathbb{I}_e = 0$ – the case of using electronic money exclusively. In this case the monopolist’s deadweight losses are likewise present; however, there is no increase in surplus and liquidity of funds for buyers. This is because buyers no longer have the ability to differentiate between the means of payment and the means of accumulation (since all funds are exchanged for electronic money for the purpose of subsequent consumption). In addition, the authors single out the monopolist’s negative externality, since he is essentially a superfluous intermediary in the exchange between buyer and producer.

Thus, the key difference of this model (and, as it turns out, its main advantage) compared to the model of ordinary private money [15] is the coexistence of two types of money and the partial division of monetary functions between them. As the analysis of the impact on social welfare showed, it is precisely the division of the functions of money that leads to the most optimal result within the model (especially for buyers). Market competition in issuance in the

money market, which represents a decentralized way of doing it, if one follows the conclusions of the model of ordinary private money [15], leads to instability of the banking system. The authors of the electronic money model do not propose a way to solve this problem; instead, they describe a centralized issuance mechanism. These two main results point to the efficiency of a two-tier monetary system.

Cryptocurrencies, as is known, were not intended to replace ordinary money but were developed as an alternative means of payment. Some assumptions of the model contradict the idea of cryptocurrencies: thus, cryptocurrencies are issued in a decentralized manner, in contrast to the centralized issuance of electronic money in the model, and electronic money cannot serve as a means of accumulation. The model can be modified to get rid of the first contradiction: if one gets rid of the monopolist firm, the deadweight losses and the surplus it creates disappear; however, owing to the specifics of cryptocurrency issuance, the costs of its mining arise. As for how to adapt to the second contradiction, it is not so easy to do. If electronic money is endowed with the function of a means of accumulation, the costs of using electronic money change, since “shoe-leather costs” arise in it (which would be insignificant owing to the ease of managing crypto-finances) along with storage costs (which may be substantial because of the volatility of the crypto market). As a result of this change, the condition for choosing between money and electronic money changes. An analogous table for cryptocurrencies would look approximately as presented in Table 2

Table 2: Impact on welfare in the cryptocurrency model. (Source: compiled by the author)

Agent	Indicator	$\mathbb{I}_c = 1$	$\mathbb{I}_e = 1$	$\mathbb{I}_e = 0$
Buyer	Liquidity of funds and transaction security	0	+	0
	Externalities	0	?	?
	Mining costs	0	-	-
Total		0	+	-
Producer	Liquidity of funds and transaction security	0	+	0
	Externalities	0	?	?
	Mining costs	0	-	-
Total		0	-	-

One can conclude that the key question in an analogous model with cryptocurrencies instead of electronic money would be the question of comparing the social losses from issuing a cryptocurrency and the potential benefit from its use.

2.4 Cryptocurrency as a type of private electronic money

Now, understanding the path of economic thought regarding private money and private electronic money, we can move on to the object of the research – cryptocurrencies. Cryptocurrencies are a private decentralized digital (electronic) currency. Following the reasoning of the Austrians, the private decentralized method of issuance is subject to market laws, and therefore the issuance of a cryptocurrency is self-regulating. In addition, as can be seen from the private electronic money model discussed earlier, the use of a cryptocurrency as a means of payment, compared with ordinary money, is distinguished by increased monetary liquidity. Let us analyze the technical process of cryptocurrency issuance.

Bitcoin is the most popular and stable cryptocurrency, but in addition to this, it is an excellent proxy generalizing most existing cryptocurrencies. Let us examine the algorithms and functioning of cryptocurrencies using it as an example. To begin with, Bitcoin is issued exclusively in a decentralized manner by users who process transactions, in the form of blocks. A block contains several BTC, and gradually the number of bitcoins in one block is halved. Thus, at the start of Bitcoin, one block contained 50 “coins,” then after three “halvings” it reached the level of 6.25 BTC per block; the next one should take place in 2024. The process of processing transactions using technical means is called “mining.” The total supply of bitcoins is limited: at present more than 19 million BTC have been issued, with a permissible issuance limit of 21 million BTC. At the same time, the per-period issuance volumes are adjusted using two features of the algorithm: a different block volume and a different “mining” difficulty. The block volume (how many bitcoins are contained in one block) that is mined for the same number of completed transactions constantly changes over time, being an endogenous indicator. The “mining” difficulty also adjusts to the environment: if bitcoin blocks are mined too quickly, the difficulty increases, bringing the time to mine one block closer to the target of 10 minutes. Thus, the algorithm of bitcoin issuance is self-regulating, which, for the most part, is what attracted attention to it.

It should be noted that mining a cryptocurrency is an incredibly energy-intensive process. This explains the fact that crypto-farms are very clearly concentrated in certain regions with low electricity costs. The direct costs of mining a cryptocurrency are the costs of computing equipment (ASIC, graphics card, or processor) and the cost of the electricity expended. The costs can also include the computation algorithm used – a faster or more energy-intensive one.

It is logical that Bitcoin and many other cryptocurrencies are “floating-rate” currencies. This means that their value depends on the supply of and demand for the cryptocurrency itself, rather than on a pegged commodity or currency, as in the case of other digital currencies:

E-Gold (a digital currency pegged to gold), eUSD (a crypto token pegged to the US dollar). At the same time, one cannot deny the importance of the impact of issuance costs on the issuance of a cryptocurrency: since the issuance of the currency is decentralized, the ratio of the cryptocurrency price to the technological and energy costs determines the decision made by each individual about issuing the cryptocurrency. Therefore it is necessary to analyze the theoretical approach to analyzing the cryptocurrency market taking these two factors into account.

Another analogy arises when comparing the ideas of Bitcoin pricing, according to the cryptocurrency algorithms set out above, and the “pricing” of money under the gold standard [1]. By analogy with the profitability of gold mines, the BTC exchange rate, as is confirmed in some studies [10], depends on the technical equipment used in “mining” and even on the price of electricity.

The next logical continuation of the research, it seems to me, is an analysis of models of demand for cryptocurrency and of cryptocurrency pricing based on costs. They are consistent with the ideas about the issuer’s guarantees and the correlation with mining costs, respectively. It should be noted that these models are precisely the key direction of scientific activity in the field of the economic analysis of cryptocurrencies.

2.5 The model of demand for cryptocurrency

First of all, it is worth discussing the model of the impact of various factors on the demand for cryptocurrencies, since the question “What should induce an individual to use money?” was asked as far back as Hayek. The most successful work, in my view, is the model constructed by M. Arias-Oliva [3]. In addition, the views of the authors of the model can be compared with the views of the authors of the model of demand for private electronic money [6].

In fact, the aim of M. Arias-Oliva’s study can be presented somewhat more broadly than simply analyzing what is a significant factor affecting the demand for cryptocurrency. The authors themselves often move from cryptocurrency to an attempt to answer the question “what affects the population’s trust in new technologies?” However, as many other authors believe ([4], [2], [6]), as well as contemporary representatives of the Austrian school, the concept is inseparable from information technologies, and an approach to them as to ordinary money, or even as to ordinary electronic money, is incorrect.

The authors put forward several hypotheses regarding the demand for cryptocurrency, drawing on research on new technologies appearing on the market:

H1: high expectations of an asset's return positively affect the intention to use it [21].

H2: expectations of the degree of ease of use of an asset positively affect the intention to use it [25].

H3: network effects positively affect the intention to use an asset [16].

H4: the presence (creation by external economic agents) of favorable conditions for using an asset positively affects the intention to use it [16].

H5: risk expectations of an asset negatively affect the intention to use it [13].

H6: financial literacy positively affects the intention to use various financial assets [12].

A PLS-SEM (Partial Least Squares – Structural Equation Modeling) econometric regression model is constructed to test the listed hypotheses. The model is tested on survey data of residents of Spain over the age of 20 for 2018. Each of the 402 respondents answered a number of questions that can be grouped into six categories corresponding to the regressors, and also stated their intentions regarding the use of cryptocurrency as an asset or a means of payment.

The results allow us to say that only two hypotheses are confirmed at the 1% significance level (hypotheses *H1*, *H4*), and three at the 10% level (hypotheses *H1*, *H2*, *H4*). The hypotheses about the positive effect of network effects and financial literacy and the negative effect of risk expectations cannot be accepted at any significance level, which is a rather unexpected result. Nevertheless, the relatively high value of R^2 and the greater-than-zero value of Q^2 (PLS Predict – see G. Shmueli, 2016 [22]) indicate that the model describes the data well and that the authors' conclusions are valid. Nevertheless, the descriptive statistics of the data show that the intentions to use cryptocurrency were, on the whole, low. I suspect that on different samples (especially from different countries with different degrees of acceptance, including legislative acceptance, of cryptocurrencies) the estimates will differ, so one should not speak of a strict rejection of hypotheses *H3*, *H5*, *H6*.

As for the comparison with the previously considered model of private electronic money [6], the willingness to use a means of payment other than the customary one depended, first of all, on a comparison of the costs of using state money and private (electronic) money. In this model, however, there is an assumption that individuals may treat cryptocurrency as an asset (which is close to reality), and not only as a means of payment. Despite all the methodological differences, the analysis of the demand for cryptocurrencies showed that in real life there is also a weighing of the “utility” and “costs” of using cryptocurrency, judging by the three significant variables

– expectations of the cryptocurrency’s return, expectations of the degree of difficulty, and the presence of the possibility of using cryptocurrency.

Therefore it will be logical to move on to a detailed discussion of the impact of costs on the cryptocurrency market and the possible economic consequences.

3 Theoretical aspects of the costs of issuing a cryptocurrency

3.1 The environmental aspect

We have already determined the structure of mining costs in the previous chapter. To recall: they directly include technological costs (the costs of computing equipment), energy costs (the costs of the electricity used in the “mining” process), and the opportunity costs of choosing a computation algorithm (for example, SHA-256 or the newer Scrypt).

The greater part of “mining” costs is, of course, made up of energy costs. Cryptocurrencies are so energy-intensive to issue that increasing the energy efficiency of altcoins has become one of the important technological directions in the context of today’s environmental agenda. So-called “green” cryptocurrencies have begun to appear and to be actively promoted – their distinguishing feature is that, on average, less electricity is required to issue them, and they also use energy obtained from renewable sources [19].

There are contemporary environmental studies that analyze the costs of issuing cryptocurrencies and put forward proposals on how to make global finance more energy-efficient.

The research article by B. Ren and B. Lucey [20] investigates the relationship between clean energy and cryptocurrencies, especially in terms of their environmental impact. The article examines the use of clean energy sources, such as solar and wind energy, in cryptocurrency mining and how this affects the overall carbon footprint of the cryptocurrency industry.

The authors argue that, although the use of clean energy in cryptocurrency mining is increasing and the energy efficiency of mining is growing, the industry still uses a significant amount of dirty energy. Relevant for my research is the authors’ observation that the main factors affecting the costs of mining a cryptocurrency are the lack of regulation in the energy industry, the high profitability of “mining,” and the limited availability of clean energy sources in some regions.

The authors suggest that encouraging the use of clean cryptocurrencies may be a way to incentivize the industry to transition toward more sustainable pricing and greater social welfare (by reducing the negative effect of mining costs, as presented in Table 2). They also recommend that legislators and regulators take a more active role in encouraging the use of clean energy in the industry and in complying with environmental standards.

The research article by L. Pham [19] and others examines the relationship between energy

prices and the value of “green” and “non-green” cryptocurrencies. The authors argue that energy pricing policy can have a significant impact on the cryptocurrency market, since it creates incentives for the use of clean energy sources in cryptocurrency mining. They identify two relationships in the cryptocurrency market: one for “green” cryptocurrencies, which have a smaller carbon footprint and whose share grows as carbon prices rise, and one for “non-green” cryptocurrencies, which have a higher carbon footprint and whose mining intensity falls as the prices of non-renewable energy rise.

The obvious takeaway of this work is, of course, the proposal to increase social welfare by encouraging the use of renewable energy sources and abandoning “dirty” cryptocurrencies. However, it should be noted that the authors point out an important effect of the impact of state regulation on decisions about cryptocurrency “mining”: in Western states, mining “green” cryptocurrencies is in fact more profitable. This is consistent with the fact that the share of “green” farms in the EU and the US is higher than in other countries. The article also examines the concept of “greenwashing” in the cryptocurrency industry, which refers to the practice of promoting a cryptocurrency as environmentally friendly without actually reducing its carbon footprint. The authors suggest that such a practice is becoming increasingly common as the demand for “green” cryptocurrencies grows.

The regulation of, and the attitude of state authorities toward, cryptocurrencies is dictated not only by financial interests and considerations of payment security. The environmental and energy aspects directly affect the decisions of particular states to ban “mining.” For example, at the end of 2021, Kazakhstan ranked second in “mining” capacity. When the legislative bodies turned their attention to crypto-farms and their high energy consumption, they began to take restrictive measures and to license “mining” income with the aim of protecting the energy industry, as a result of which Kazakhstan’s share of global cryptocurrency mining capacity significantly decreased. Not all countries try to limit the energy consumption of crypto-farms. In the PRC, a decision was made to completely ban operations with cryptocurrencies and their mining. Incidentally, owing to cheap electricity rates, China’s share of total Bitcoin “mining” in mid-2021 was 44%.

3.2 The cost-of-production model of cryptocurrency pricing

One of the popular approaches to analyzing the cryptocurrency market is modeling cryptocurrency pricing based on the costs of its issuance. The approach in these models is more “technological” than in the models of demand for cryptocurrency: the change in price is assumed to be explained not by demand shocks (the previous demand-analysis model gives, in principle,

an exhaustive explanation [3]), but by a change in such factors as the price of electricity, the equipment used (hardware), and the “mining” algorithms (software). Such an approach is indeed valid, since one can easily formulate the costs of issuing bitcoin, which will consist, for the most part, of the cost of the listed factors.

Let us consider the study by A.S. Hayes [10]. The authors consider the price of a cryptocurrency in terms of variables that are in one way or another embedded in the cryptocurrency protocol. Let me list them:

1. *CP* – computational power – the computing power involved in mining the altcoin, Giga-Hashes/s.
2. *CPM* – coins per min – the reward for finding a block (initially, for example, the reward for a Bitcoin block was originally 50 BTC; at the time of the study, 12.5 BTC), divided by the average time to find one block (for BTC, 10 minutes), number of altcoins/min.
3. *mined%* – mined coins share – the share of the cryptocurrency that can still be issued (for BTC, for example, 9.5% of the possible supply remains to be issued), %.
4. *ALGO* – algorithm – the algorithm used for finding blocks, 1 – Script is the “mining” algorithm used.
5. *D* – days since – the number of days elapsed since the launch of the altcoin.

PRICE – altcoin price – the variable under study, the value of the altcoin in dollars.

The authors use mining power instead of aggregate costs, pointing to the positive correlation between these two quantities. It is possible that the mining power of a cryptocurrency and the price of the cryptocurrency affect each other bidirectionally, since, on the one hand, an increase in power increases the costs of finding a block as a result of the difficulty retarget, and on the other hand, a high price stimulates the attraction of new computing power. Granger causality testing shows that, most likely, the mining power of a cryptocurrency affects the price one-directionally, so the authors formulate the first hypothesis:

H1: the aggregate mining power positively affects the price of the corresponding altcoins.

By the logic of the law of diminishing marginal utility, the more available a cryptocurrency is, the less valuable it is. Within the framework of the listed variables, the hypothesis can be formulated as follows:

H2: the number of units of an altcoin issued per unit of time negatively affects the price of the corresponding altcoin.

The law of scarcity can be applied to cryptocurrencies because of the exogenous limit on their supply:

H3: the ratio of the mass of cryptocurrency already issued to that which is yet to be issued positively affects the price of the corresponding altcoin.

Bitcoin is mined using the SHA-256d algorithm; subsequently, many other substitute algorithms were released for “younger” cryptocurrencies. Most altcoins are mined using the Scrypt algorithm, which is considered more efficient. Hence the corresponding hypothesis arises:

H4: the price of altcoins mined on the basis of the Scrypt algorithm is relatively higher than the price of cryptocurrency mined on the basis of SHA-256d.

The following hypothesis echoes Hayek’s assumptions about the demand for private money depending on trust in its issuer [9]:

H5: the time of an altcoin’s presence on the market positively affects the price of the corresponding altcoin.

To test the hypotheses, an OLS model was constructed:

$$\ln(PRICE) = \beta_0 + \beta_1 \ln(CP) + \beta_2 \ln(CPM) + \beta_3 \text{mined\%} + \beta_4 ALGO + \beta_5 D + \epsilon \quad (4)$$

4: Regression model of cryptocurrency value

The model is tested on open data on the 66 most popular altcoins as of 2014 (unbalanced panel data).

At the 1% significance level, hypotheses *H1*, *H2*, and *H4* are confirmed. The hypotheses about the positive effect of the “scarcity” of altcoins and of the time of an altcoin’s presence on the market cannot be accepted at any significance level. There are grounds to doubt these conclusions, since it makes sense to also consider models with fixed and random effects (with and without time effects), because an ordinary pooled regression often gives incorrect estimates owing to the complexity of the assumptions. It is also desirable to conduct a test for endogeneity in the model in case the bidirectional influence of “mining” power and the bitcoin price does

exist, or there is a high correlation between CP and CPM if the difficulty retarget has a certain lag for some altcoins. In that case, it is necessary to select an instrument for “mining” power, which is not so easy to do.

Nevertheless, the results obtained by the authors are valuable, since such a model set the vector of development for many econometric models of cryptocurrency price analysis and gave rise to adjacent models of cryptocurrency issuance costs. It is precisely on the basis of this model that I constructed my modified version of the model for the empirical analysis of the impact of “mining” costs on the cryptocurrency market.

3.3 The endogenous model of the cryptocurrency market

Another interesting approach is to consider the markets for cryptocurrencies and the markets for the materials needed to issue them as interrelated. The notable article by D. Okorie [18] examines the relationship between the markets for Bitcoin, Ethereum, and electricity in the US, the PRC, and Japan.

The author of the article puts forward hypotheses about the positive effects of each of the variables under study on one another: the returns and trading volumes of Bitcoin and Ethereum, and the demand for electricity in the US, the PRC, and Japan. To test the hypotheses, the authors resort to the VAR (Vector Autoregression) model. The data are taken from August 7, 2015 to December 31, 2019.

The author proves the validity of all the hypotheses put forward, and also notes the difference between the electricity markets in the US/Japan and the PRC. They explain that, with the trend in “miner” activity computed by the author persisting, CO₂ emissions into the atmosphere as a result of the energy expended on issuing cryptocurrencies will constantly increase over time, since a large share of the energy consumed is energy obtained from non-renewable sources in the PRC. It should be noted that, after the ban on operations with cryptocurrencies in the PRC, the concerns of the author of the study about a substantial growth in CO₂ emissions become less well-founded, since the lion’s share of “mining” is taken by the US, where “green” energy is developed.

4 An empirical model of the impact of costs on the cryptocurrency market

4.1 Formulation of the hypotheses

Drawing on the theoretical foundation of “mining” costs set out above, it is possible to conduct a quantitative study and estimate the impact of costs on the cryptocurrency market. However, unlike the empirical works related to the cryptocurrency market that I have listed, my model will be needed to test different hypotheses.

In the article by A.S. Hayes [10], the idea is expressed that “mining” power affects the price of the cryptocurrency being issued because the mining power contains the aggregate costs of issuing the cryptocurrency, which, as the author proves, participate in the formation of the “fair” price of the cryptocurrency. Moreover, the author argues that “mining” power affects the price one-directionally, referring to the Granger causality test. In the article by D. Okorie [18], a bidirectional relationship is described between the costs of issuing cryptocurrencies (the demand for electricity) and the price of the cryptocurrency. I would like to call into question the one-directionality of the influence of price and costs, since the environmental articles [19] [20] and common sense suggest that the price of a cryptocurrency positively affects “mining” power because it raises the instantaneous profitability of mining the cryptocurrency and thereby creates incentives for “mining.”

It should also be said that the price of a cryptocurrency most likely has a long-run impact on aggregate power, since it motivates new users to buy mining equipment and become one of the issuers of the cryptocurrency. On the other hand, price shocks have a short-run effect as well, since, depending on the profitability of “mining,” issuers may decide to activate and deactivate “mining” farms within a few days. At the same time, one can hardly speak of a long-run effect of aggregate power on the price of a cryptocurrency, since the “fair” price of a cryptocurrency should be formed depending on the costs expended on its issuance over the last few days.

Therefore the following hypotheses can be formulated:

H1: the price of a cryptocurrency has a significant positive short-run impact on the aggregate costs of its mining.

H2: the price of a cryptocurrency has a significant positive long-run impact on the aggregate costs of its mining.

H3: the aggregate costs of mining a cryptocurrency have a significant positive short-run impact on the price of the cryptocurrency.

H4: the aggregate costs of mining a cryptocurrency have a significant positive long-run impact on the price of the cryptocurrency.

4.2 Data

To test the hypotheses, data on the price of a cryptocurrency are needed, as well as information on the aggregate costs of “mining.” I use data on the aggregate costs in the form of the daily electricity consumption of crypto-farms. Quantitative estimates on this dataset will make it possible both to test the validity of the hypotheses put forward and to take a step forward in environmental research on the impact of energy costs.

I prove the hypotheses on the basis of daily Bitcoin data for the period from January 1, 2015 to December 4, 2021. Data on the daily price of Bitcoin were obtained from the CoinMarketCap website [26]. Data on the electricity consumption of Bitcoin crypto-farms were calculated using the Bitcoin electricity consumption index provided by the University of Cambridge [27], with respect to daily electricity consumption.

The data represent two time series: price (the price of the cryptocurrency in US dollars per unit of BTC) and energy (the electricity consumption of crypto-farms in GW). The descriptive statistics of the time series are presented in Table 3.

Table 3: Descriptive statistics. (Source: compiled by the author)

Statistics	price	energy
Mean	10774.02	40948.6
Median	6338	36800
SE	308.06	677.74
Min	162.0	1740.0
Max	67559.0	132000.0
Skew	2.05	0.66
Observations	2530	2530
Unit	\$ per BTC	GW

To test the necessary condition for a bidirectional cause-and-effect relationship, I will conduct a Granger causality test. I will take the number of lags for each test equal to 7 (that is, I will test whether there is a significant influence of the value of a variable from the 7 preceding periods on the value of the second variable in the current period). For each of the two tests (testing the significant influence of price on energy and of energy on price), the p-value turned out to be significantly less than 0.01, so at the 1% significance level one can assert that there is bidirectional Granger causality between the series.

Next, I will use the ITCV (impact threshold of a confounding variable) method, presented in the article by K. Frank [8], to confirm or refute the absence of the influence of the cryptocurrency price on the “mining” costs claimed in the study by A. Hayes [10]. The idea is to calculate, for a linear model, a threshold indicator. If the coefficient on a regressor is above the threshold, then its influence on the sample is recognized as significant. The value of the coefficient on the variable price and the value of the calculated threshold in the linear model on the sample I use are presented in Figure 2. On the basis of this test, one can conclude that the influence of price is significant.

Figure 3 presents the time series listed above. One can notice that their dynamics are similar, but electricity consumption reacts to a change in price with a certain lag: this is especially noticeable in mid-2021.

It should be stipulated that, within the framework of daily data, I will consider as a short-run effect the effect in the first two weeks after a change in a variable. The effect after two weeks I treat as long-run.

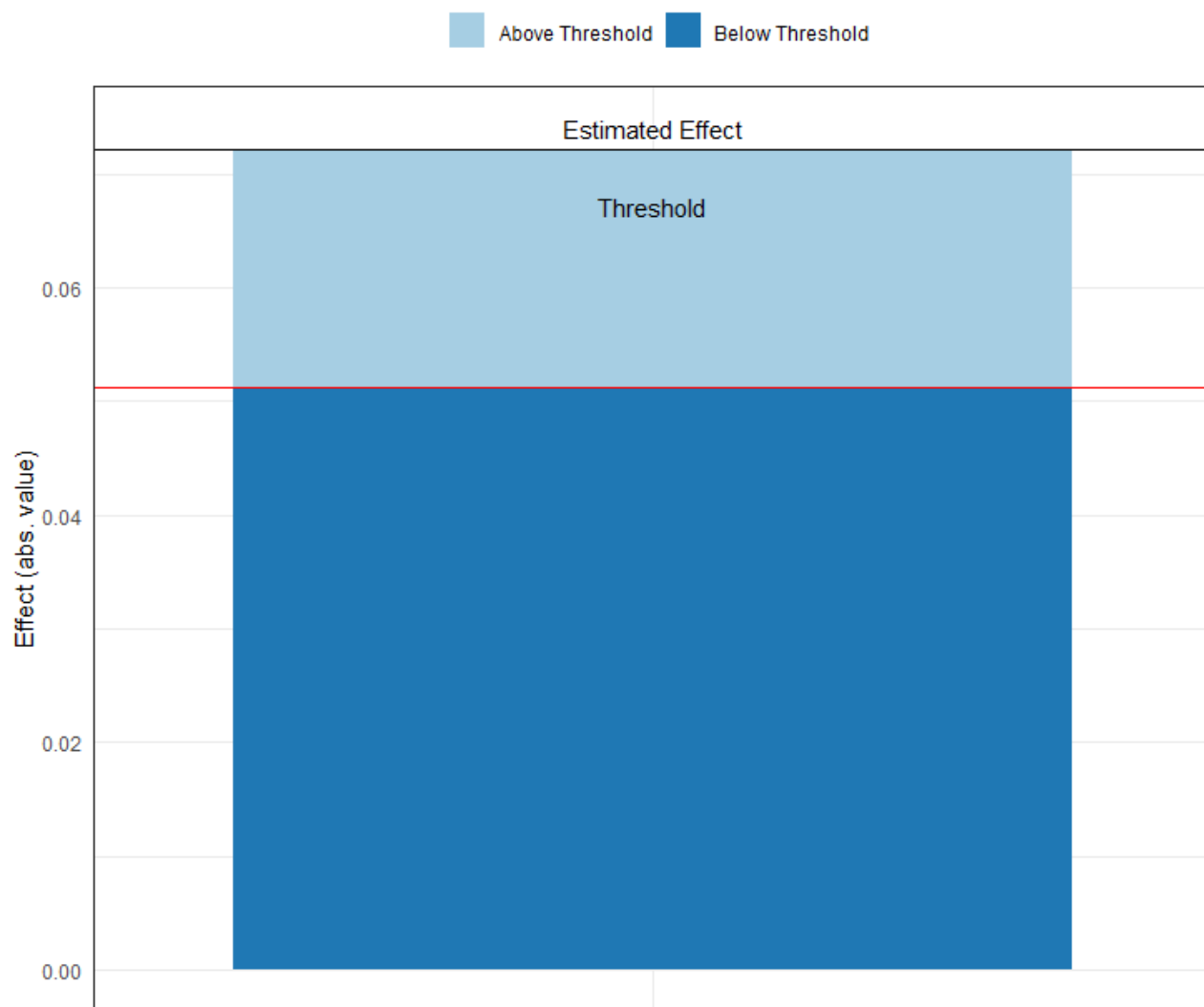


Figure 2: ITCV test for the variable price. (Source: compiled by the author)

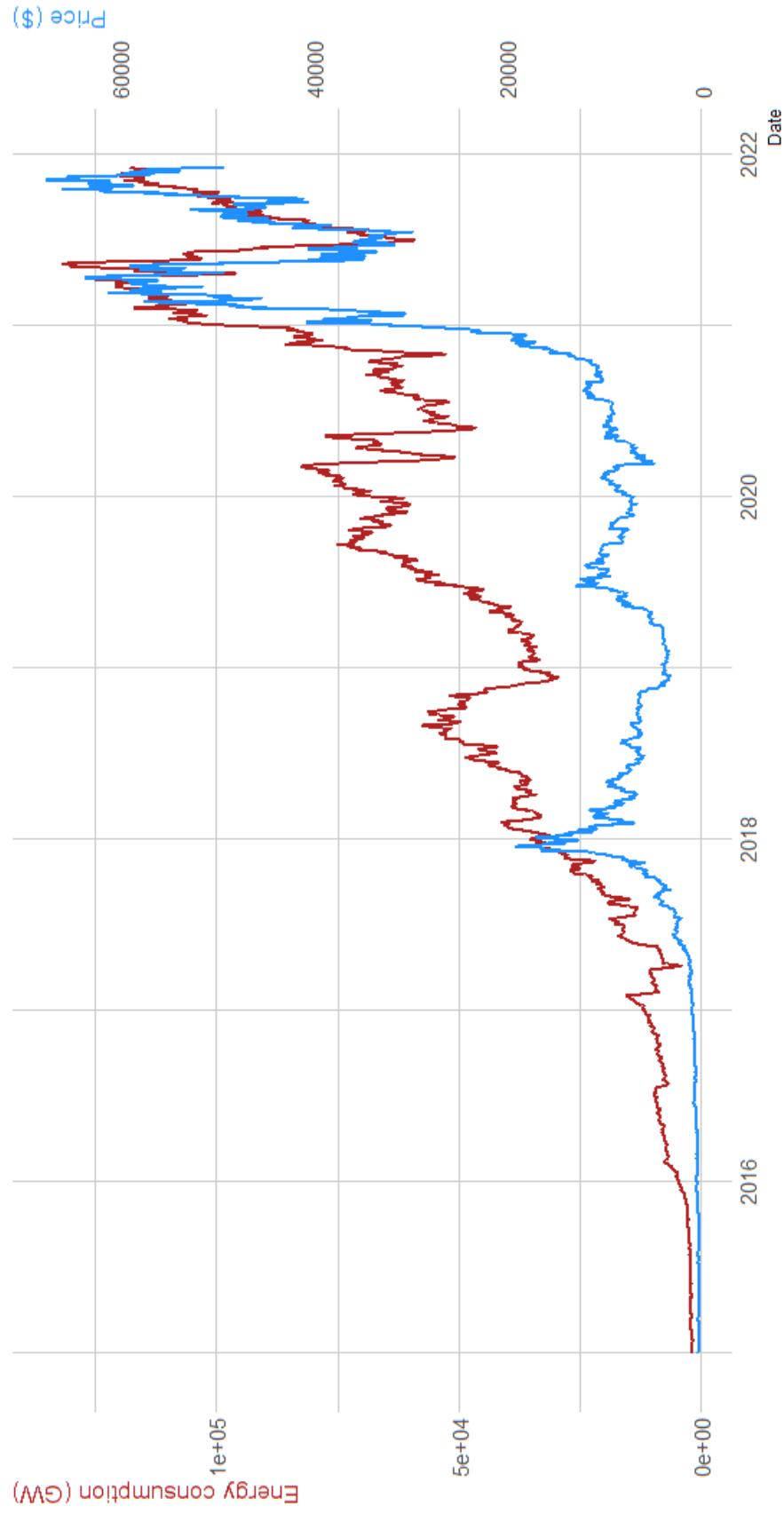


Figure 3: Dynamics of the Bitcoin price and the electricity consumption of Bitcoin crypto-farms. (Source: compiled by the author)

4.3 Model selection

Since I assume a bidirectional relationship, it is necessary to use a vector regression model owing to endogeneity. I make a choice between a vector error correction model (VECM) for cointegrated non-stationary time series and a structural vector autoregression (SVAR) model. In order to decide which model is better suited to this case, it is necessary to test the stationarity of the time series and the order of cointegration between them.

From the graphs of the time series it can be seen that they are non-stationary. Let us verify this using the augmented Dickey-Fuller test. The p-values of the tests (0.67 for price and 0.04 for energy) indicate that neither series is stationary at the 1% significance level.

Let us test for the presence of cointegration between the two time series. For this I will use the Johansen test taking into account 14 lags and a linear trend. The results are presented in Table 4.

Table 4: Johansen cointegration test. (Source: compiled by the author)

Level of cointegration	test	0.1	0.05	0.01
$r \leq 1$	4.03	10.49	12.25	16.26
$r = 0$	21.35	22.76	25.32	30.45

The Johansen test shows that at no significance level can the hypothesis that the order of cointegration between the time series is greater than zero be accepted. This means that the use of a vector error correction model (VECM) in this case is not justified.

I make the choice in favor of a structural vector autoregression. Let us check whether the first differences of the time series are stationary – this condition is necessary for applying the SVAR(p) model. The augmented Dickey-Fuller test shows that both series are stationary in first differences at the 1% significance level. The graphs of the series are presented in Figure 4.

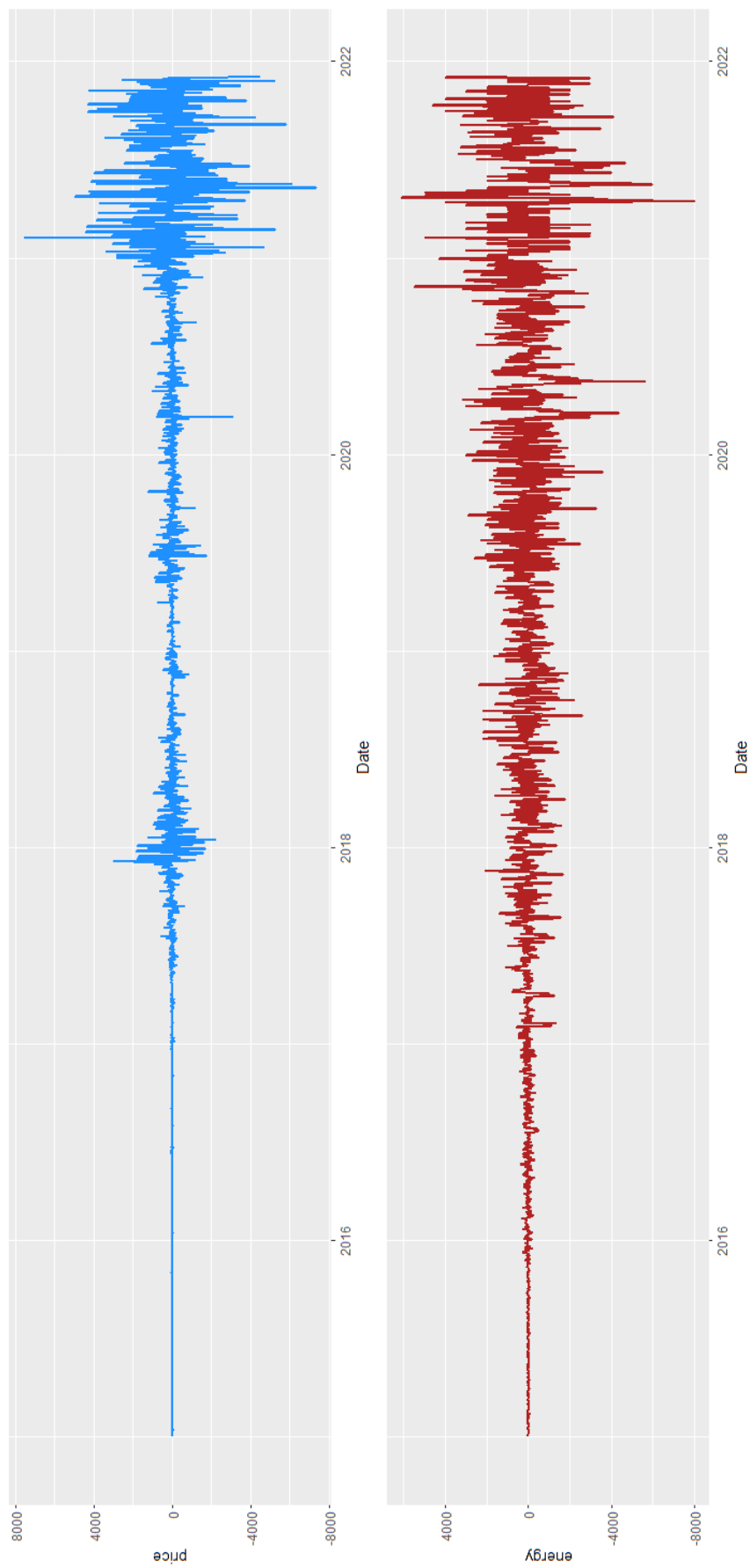


Figure 4: Dynamics of the increments of the Bitcoin price and the electricity consumption of Bitcoin crypto-farms. (Source: compiled by the author)

4.4 Model specification

Thus, the SVAR(p) in the two-dimensional case will be defined by the following simultaneously estimated equations 5. $Price_t$ and $Energy_t$ are the observations of the corresponding time series at time t . p is the number of lags in the SVAR model (the order of the SVAR model), γ and b are the estimated coefficients, ϵ are the shocks.

$$\begin{aligned} price_t = & \gamma_{100} - b_{12} * energy_t + \gamma_{111} * price_{t-1} + \gamma_{121} * energy_{t-1} + \dots + \\ & + \gamma_{11p} * price_{t-p} + \gamma_{12p} * energy_{t-p} + \epsilon_{1t} \end{aligned} \quad (5)$$

$$\begin{aligned} energy_t = & \gamma_{200} - b_{21} * price_t + \gamma_{211} * price_{t-1} + \gamma_{221} * energy_{t-1} + \dots + \\ & + \gamma_{21p} * price_{t-p} + \gamma_{22p} * energy_{t-p} + \epsilon_{2t} \end{aligned}$$

5: The SVAR(p) model in the form of two ARDL equations.

Let us write the structural vector autoregression equation in structural form 6 and reduced form 7.

$$\begin{aligned} B * y_t = & \gamma_0 + \Gamma_1 * y_{t-1} + \dots + \Gamma_p * y_{t-p} + \epsilon_t, & B = \begin{pmatrix} 1 & b_{12} \\ b_{21} & 1 \end{pmatrix} \\ y_{t-p} = & \begin{pmatrix} price_{t-p} \\ energy_{t-p} \end{pmatrix} & \gamma_0 = \begin{pmatrix} \gamma_{100} \\ \gamma_{200} \end{pmatrix} & \Gamma_p = \begin{pmatrix} \gamma_{11p} & \gamma_{12p} \\ \gamma_{21p} & \gamma_{22p} \end{pmatrix} & \epsilon_t = \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix} \end{aligned} \quad (6)$$

6: The SVAR(p) model in structural form

$$y_t = \alpha_0 + A_1 * y_{t-1} + \dots + A_p * y_{t-p} + u_t, \quad (7)$$

$$\alpha_0 = B^{-1}\gamma_0 \quad A_p = B^{-1}\Gamma_p \quad u_t = B^{-1}\epsilon_t$$

7: The SVAR(p) model in reduced form

I will select the order of the VAR(p) model by comparing the values of the Schwarz information criteria of models with different numbers of lags. The smallest criterion belongs to the model with 18 lags: it equals 26.99. The heteroscedasticity test according to the idea proposed

by Engle [7], as well as the Breusch-Godfrey test for autocorrelation, show that the model contains both heteroscedasticity and autocorrelation in the residuals. Therefore, to estimate the VAR model, I use robust standard errors (vcovHAC – Heteroscedasticity and Autocorrelation Consistent).

To test the hypotheses of interest, it is necessary to estimate the impact of exogenous structural shocks – to derive the impulse response functions. For this, let us introduce an identifying restriction. It is logical to assume that an energy consumption shock has no instantaneous impact on the price of the cryptocurrency, since information about the amount of mining becomes known with a certain lag, whereas information about the Bitcoin price is available at any moment, and issuers can turn equipment on and off depending on profitability.

$$b_{21} = 0, \quad \longrightarrow \quad B = \begin{pmatrix} 1 & b_{12} \\ 0 & 1 \end{pmatrix} \quad (8)$$

8: The identifying restriction in the SVAR model

4.5 Model results

Now we can construct the graphs of the impulse response functions. Figures 5, 6, 7, 8 present the accumulated effects of changes in the exogenous shocks of the variables by 1 standard deviation on the values of the variables under study over the following month. The red dashed lines denote the confidence intervals at the 5% significance level. In Figure 5 one can notice that the effect of a positive electricity consumption shock significantly positively affects the Bitcoin price up to the 18th day, after which the effect ceases to be significant. In Figure 6 it can be seen that the accumulated effect of a positive Bitcoin price shock significantly positively affects electricity consumption throughout the entire month, and the magnitude of the accumulated effect stabilizes at approximately day 12.

Thus, one can say that hypotheses $H1, H2, H3$ are confirmed at the 5% significance level, and hypothesis $H4$ cannot be accepted at the chosen significance level, since the effect of an exogenous electricity consumption shock on the Bitcoin price after two weeks is insignificant.

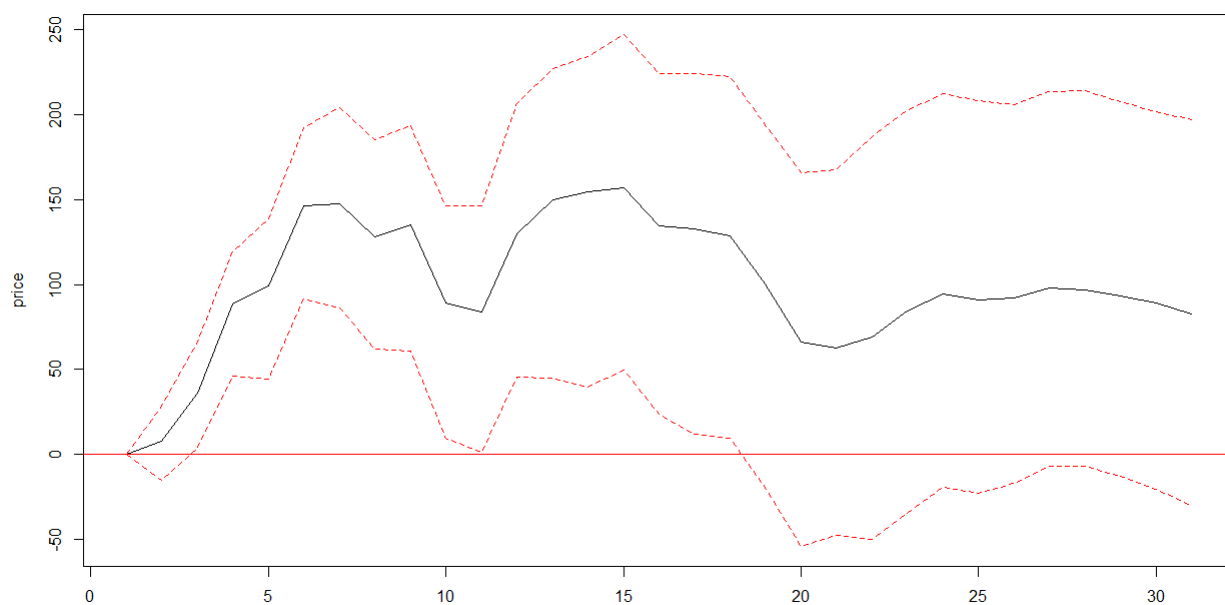


Figure 5: Accumulated impulse response function of price to an exogenous energy consumption shock. (Source: compiled by the author)

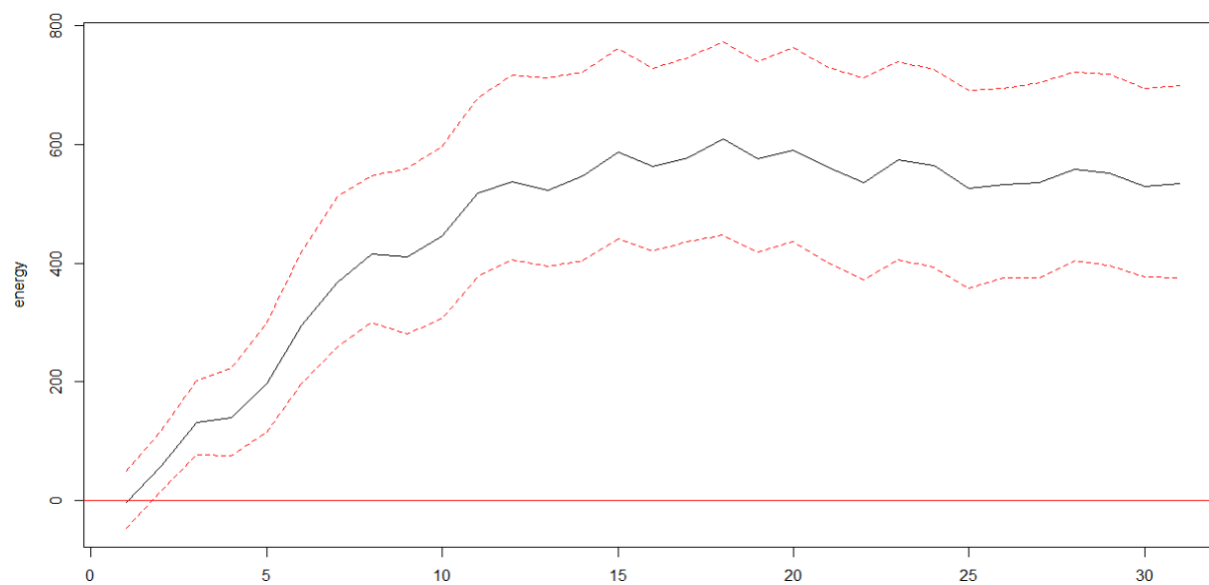


Figure 6: Accumulated impulse response function of energy consumption to an exogenous price shock. (Source: compiled by the author)

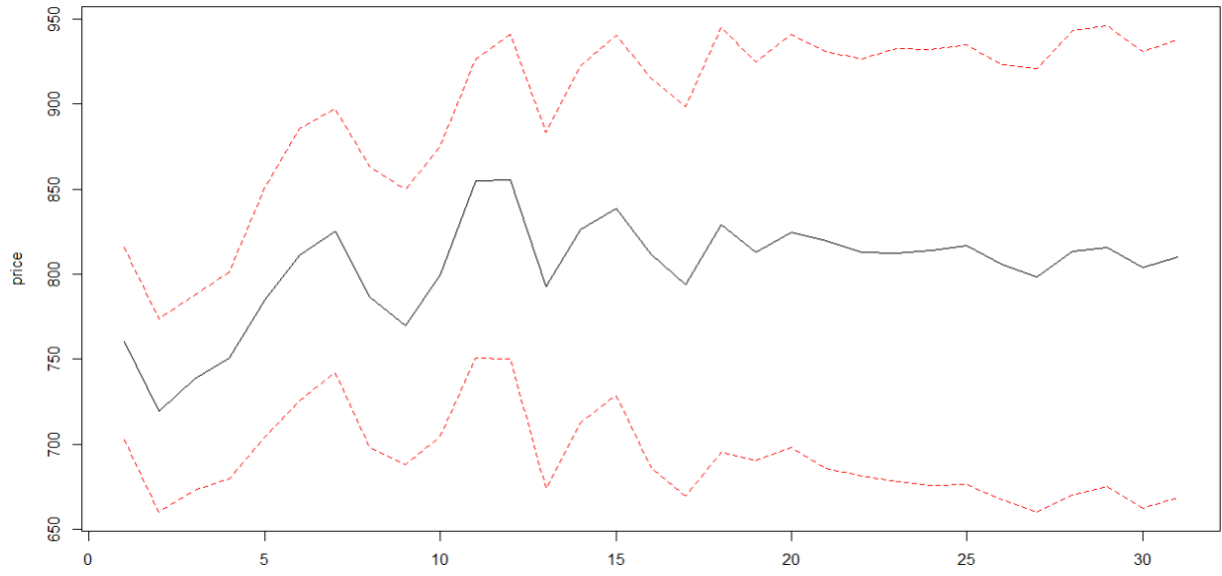


Figure 7: Accumulated impulse response function of price to an exogenous price shock. (Source: compiled by the author)

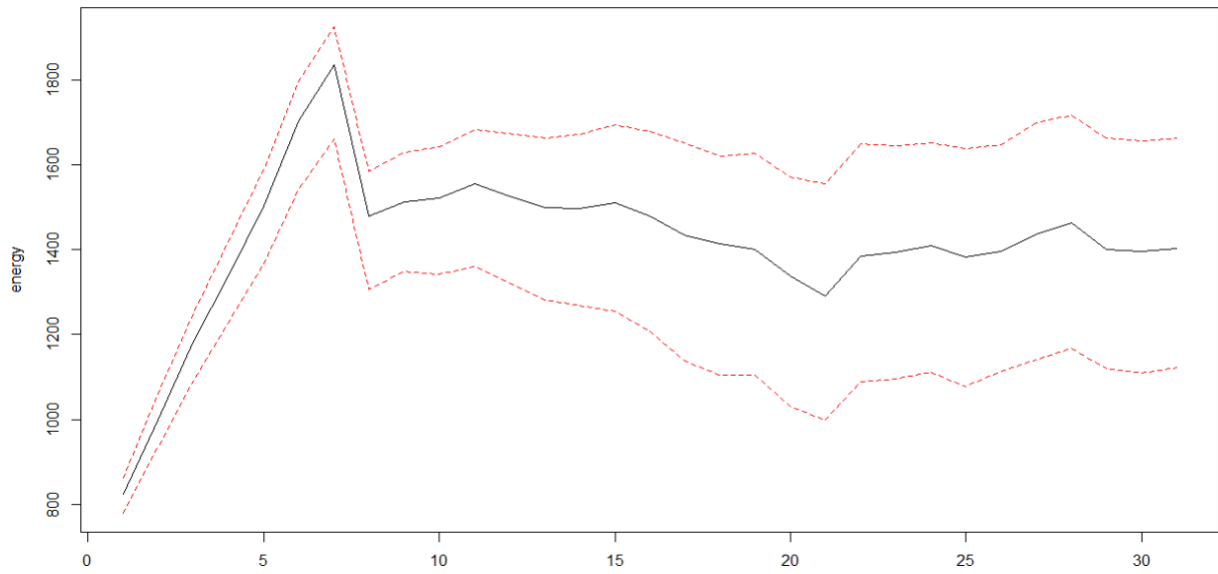


Figure 8: Accumulated impulse response function of energy consumption to an exogenous energy consumption shock. (Source: compiled by the author)

5 Conclusion

In the course of the research, I managed to single out the main theoretical differences between the cryptocurrency market and the markets for private electronic money, and to single out the most important components in the structure of the aggregate costs of issuing a cryptocurrency. On the basis of the collected data, an SVAR model was estimated, which made it possible to test the empirical hypotheses I put forward.

Relying on the results of the model, one can confidently confirm the presence of a significant positive bidirectional short-run relationship between the aggregate costs of issuing a cryptocurrency and the price of the cryptocurrency, as well as a one-directional impact of the cryptocurrency price on the aggregate “mining” costs. Thus, the first three hypotheses I put forward are confirmed and the last one is rejected, which corresponds to expectations.

The presence of a long-run effect of the cryptocurrency price on costs makes it possible to make the unfavorable forecast that, with a positive trend in the value of cryptocurrencies, energy consumption will grow as a result of an increase in the number of crypto-farms and in the aggregate “mining” capacity. And a positive trend in the cryptocurrency price is guaranteed by many financial analysts owing to the characteristics of its issuance. Since cryptocurrencies are subject to the law of scarcity, as the volume of the mined block decreases, the value of the cryptocurrency will rise. This may lead to an environmental catastrophe because of the constantly increasing energy consumption.

A solution to this problem may be the timely improvement of the energy efficiency of “mining.” New algorithms or new, more energy-efficient equipment are capable of reducing the aggregate electricity consumption of crypto-farms, and therefore the aggregate costs of mining.

References

- [1] N.D. Kondratiev, D.I. Oparin, “Large Cycles of Conjuncture: Reports and Their Discussion at the Institute of Economics,” 1928.
- [2] F.M. Ametrano “Hayek Money: The Cryptocurrency Price Stability Solution”, 2016.
<https://dx.doi.org/10.2139/ssrn.2425270>
- [3] M. Arias-Oliva, J. Pelegrin-Borondo, G. Matias-Clavero “Variables Influencing Cryptocurrency Use: A Technology Acceptance Model in Spain”, *Front. Psychol.*, 2019.
<https://doi.org/10.3389/fpsyg.2019.00475>
- [4] D.S. Bas “Hayek and the cryptocurrency revolution”, *Iberian Journal of the History of Economic Thought*, 2019.
<http://dx.doi.org/10.5209/ijhe.69403>
- [5] R. de O. Cavalcanti, A. Erosa, T. Temzelides “Private Money and Reserve Management in a Random-Matching Model”, *Journal of Political Economy* Vol. 107, 1999.
<https://doi.org/10.1086/250085>
- [6] J. Chiu, T. Wong “E-money: Efficiency, stability and optimal policy”, Bank of Canada Working Paper, No. 2014-16, 2014.
- [7] R.F. Engle “Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation”, *Econometrica*, 1982.
- [8] K.A. Frank “What Would It Take to Change an Inference? Using Rubin’s Causal Model to Interpret the Robustness of Causal Inferences”, *Educational Evaluation and Policy Analysis* Vol. 35, 2013.
<https://doi.org/10.3102/0162373713493129>
- [9] F. Hayek “Denationalisation of Money: An Analysis of the Theory and Practice of Concurrent Currencies”, London: Institute of Economic Affairs, 1976.
- [10] A.S. Hayes “Cryptocurrency value formation: An empirical study leading to a cost of production model for valuing bitcoin”, *Telematics and Informatics* Vol. 34, 2017.
<http://dx.doi.org/10.1016/j.tele.2016.05.005>
- [11] L. Kilian “Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market”, *American Economic Review* Vol. 99, 2009.
<https://www.aeaweb.org/articles?id=10.1257/aer.99.3.1053>

- [12] L.T. Lam “The association between financial literacy and problematic internet shopping in a multinational sample”, *Addict. Behav. Rep.* 6, 2017.
<https://doi.org/10.1016/j.abrep.2017.10.002>
- [13] J. Mendoza-Tello “Social commerce as a driver to enhance trust and intention to use cryptocurrencies for electronic payments”, *IEEE Access* 6, 2018.
<https://doi.org/10.1109/ACCESS.2018.2869359>
- [14] H. Minsky “Stabilizing an Unstable Economy”, 1986.
- [15] C. Monnet, D.R. Sanches “Private Money and Banking Regulation”, *Journal of Money, Credit and Banking*, Vol. 47, 2015.
- [16] Y. Moon, J. Hwang “Crowdfunding as an alternative means for funding sustainable appropriate technology: acceptance determinants of backers”, *Sustainability*, 2018.
<https://doi.org/10.3390/su10051456>
- [17] S. Nakamoto “Bitcoin: A Peer-to-Peer Electronic Cash System”, 2008.
<https://bitcoin.org/bitcoin.pdf>
- [18] D.I. Okorie “A network analysis of electricity demand and the cryptocurrency markets”, *International Journal of Finance and Economics* Vol.26, 2020.
<https://doi.org/10.1002/ijfe.1952>
- [19] L. Pham et al. “A tale of two tails among carbon prices, green and non-green cryptocurrencies”, *International Review of Financial Analysis* Vol. 182, 2022.
<https://doi.org/10.1016/j.irfa.2022.102139>
- [20] B. Ren, B. Lucey “A clean, green haven?—Examining the relationship between clean energy, clean and dirty cryptocurrencies”, *Energy Economics* Vol. 109, 2022.
<https://doi.org/10.1016/j.eneco.2022.105951>
- [21] J.A. Sanchez-Torres “E-banking in Colombia: factors favouring its acceptance, online trust and government support”, *Int. J. Bank Mark* 36, 2018.
<https://doi.org/10.1108/IJBM-10-2016-0145>
- [22] G. Shmueli “The elephant in the room: predictive performance of PLS models”, *J. Bus. Res.* 69, 2016.
<https://doi.org/10.1016/j.jbusres.2016.03.049>

- [23] B.D. Smith, W.E. Weber “Private Money Creation and the Suffolk Banking System”, Journal of Money, Credit and Banking Vol. 31, 1999.
<https://doi.org/10.2307/2601080>
- [24] H. Thornton “An Enquiry into the Nature and Effects of the Paper Credit of Great Britain”, 1802
- [25] V. Venkatesh “User acceptance of information technology: toward a unified view”, MIS Q. 27, 2003.
<https://doi.org/10.2307/30036540>
- [26] CoinMarketCap.
<https://coinmarketcap.com/>
- [27] Cambridge Bitcoin Electricity Consumption Index.
<https://ccaf.io/cbnsi/cbeci>